

Department of Computer Science, Faculty of ICT, Gujarat Vidyapith, Ahmedabad-380009

Program: MSCIT Semester: I Academic Year: 2026-27

Subject: 266010344011-Linux System Administration

Subject Teacher: Kamallesh V. Salunke

Student No.: _____ **Name:** _____

Practical No.	Lab Activity	Date Assigned	Date submission	Sign	Remark
1	Write a script to determine whether a number is positive, negative, zero, even, or odd.	14-8-26	19-8-26		
2	Write a Bash script to find the largest among three numbers.	14-8-26	19-8-26		
3	Write a shell script to calculate the sum of the first n natural numbers (1 + 2 + 3 + ... + n), where the value of n is entered by the user.	14-8-26	19-8-26		
4	Develop a Bash calculator that accepts two numbers and performs addition, subtraction, multiplication, division, and modulus operations using separate Bash functions for each arithmetic operation.	14-8-26	20-8-26		
5	Write a shell script to display the calendar for the month specified by the user as a command-line argument.	14-8-26	20-8-26		
6	Write a shell script to generate and display the following output using the system date and time: Mon Aug 24 09:30:31 EST 1998 Date is: 24-08-98 Date is: 08/24/98 Time is: 09 - 30 - 31 Time is: 09:30:31 No. of days elapsed in this year is: 234 Serial no. of day (of week) is....: 31	14-8-26	20-8-26		

	Day and Date is.....: Tue Aug 24, 1998 Time in the form AM/PM: 09:30:31 AM Good Morning The script should use appropriate Linux date/time commands and formatting options to produce each of the above outputs.				
7	Demonstrate standard input, output, and error redirection using `>`, `>>`, `<`, and `2>`.				
8	Write a shell script which concatenates all given files into a single file. Put filename before every file. e.g. *** filename : xyz ***				
9	Write a shell script to display all ASCII/text files from a specified directory. If no directory is provided as a command-line argument, the program should display the ASCII/text files from the current directory.				
10	Write a script that counts files according to their extensions.				
11	Write a Bash script to recursively search for a specified directory and display all files whose size is zero bytes				
12	Write a shell script which recursively delete *.obj , *.lst or all files with length of zero bytes				
13	Write a Bash script to accept a directory and filename from the user and recursively search for the specified file using the `find` command.				
14	Write script that accepts a filename and display whether it exists, its type, size, permissions, owner, and modification time.				
15	Write a script to determine whether a specified path is a file, directory, symbolic link, or does not exist.				
16	Write a shell script to display a birthday message for all currently logged-in users whose birthday matches the system date. The users' birthdate details are stored in a file in the following format:				

	username:login_name:birthdate Kamalesh Salunke:kamalesh:28-12 Vyjay:vijay:02-03 The script should read the file, compare each user's birthdate with the current system date, and send an appropriate birthday message for matching users who are currently logged in.				
17	Write a menu-driven Bash shell script to search and manage files in a Linux system using the find command. The script should provide the following menu options: 1. Find files based on filename. 2. Find files based on file extension. 3. Find files based on file size. 4. 18Find files based on file ownership/user. 5. Find files based on file permissions. 6. Find files modified within a specified number of days. 7. Find files not modified for a specified number of days. 8. Find files modified between two specified dates. 9. Find files satisfying multiple criteria such as filename, owner, permissions, and size. 10. Count and display the total number of matching files. 11. Exit the program. The script should use a loop and case statement to implement the menu and should validate user input wherever appropriate.				
18	Write a Bash script to create a compressed backup of a specified directory using the `tar` command. The backup filename should contain the current date and time.				
19	Display the Linux distribution, kernel version, hostname, CPU architecture, memory information, and system uptime.				
20	Create a script that reports: CPU usage, Memory usage, Disk usage, Uptime, Logged-in users, Running processes				

21	Write a Bash script to accept a username, check whether the user already exists, and create the user if the username is not already present.				
22	Write a Bash script to accept a Linux username and display all groups to which that user belongs.				
23	Write a menu-driven Bash shell script to perform common Linux user management operations. The script should provide the following menu options: 1. Create a new user 2. Delete an existing user 3. Modify user information 4. Set or change user password 5. Lock a user account 6. Unlock a user account 7. Display user information 8. Display the groups to which a user belongs 9. Display all system users 10. Check whether a user exists 11. Display currently logged-in users 12. Change a user's login shell 13. Change a user's home directory 14. Set account expiration date 15. Display account aging information 16. Remove a user along with their home directory 17. Exit Requirements • Use a menu-driven interface using case. • Use a loop so that the menu is displayed repeatedly until the user selects Exit. • Validate whether the specified user exists before performing operations. • Display appropriate success and error messages. • Check whether the script is executed with root/sudo privileges before performing administrative operations.	14-8-26	26-8-26		

	• Use standard Linux commands such as: useradd, userdel, usermod , passwd, id, groups, chage, who, getent • Ask for confirmation before deleting a user. • Handle invalid menu choices appropriately.				
24	Write a script that checks whether specified services are running and reports their status.				
25	Develop an interactive Bash menu that provides options for displaying system information, disk usage, memory usage, logged-in users, running processes, listening ports, failed services, date and time, and an option to exit the program.				
26	Write a Bash script to check the status of commonly used Linux services such as SSH, cron, NetworkManager, and firewalld, and display whether each service is running or stopped.				
27	Create a script that generates an alert when disk usage exceeds a specified percentage.				
28	Write a Bash script that accepts a hostname or IP address from the user and checks network connectivity using the `ping` command.				
29	Write a Bash script to display all TCP and UDP ports currently listening on the Linux system using the `ss` command.				

Grade	A	B	C	D	E	F	S
Marks	18 - 20	14-17	11-13	8-10	5-7	1-4	Not done